Lab 3 Report: EKF Localization

Jerry Cheng
Dept. of Mechanical and Aerospace Engineering
New York University
qc1007@nyu.edu
Kaiyi Huang
Dept. of Mechanical and Aerospace Engineering
New York University
kh4195@nyu.edu

***Abstract*—This report presents an Extended Kalman Filter (EKF) for the planar 2D localization of an Ackermann-steered mobile robot. To mitigate open-loop dead reckoning drift, the filter fuses proprioceptive data from the motion model with absolute pose estimation derived from an overhead camera and an onboard AprilTag. Accurate global pose tracking is established by calibrating the camera intrinsics and dynamically defining the extrinsic world frame relative to the robot's initial AprilTag pose. Offline evaluations demonstrate rapid convergence from initial state errors of up to $2.0\,\mathrm{m}$ within $0.5\,\mathrm{s}$ and reliable uncertainty propagation during $0.5\,\mathrm{s}$ visual occlusions. Finally, real-time deployment validates the filter's ability to continuously bound odometry drift, establishing a robust state estimation foundation for future closed-loop Proportional-Derivative (PD) autonomous navigation.**

I. Introduction

Reliable state estimation is a fundamental for autonomous mobile robotics. While kinematic motion models, such as the Ackermann steering model derived in lab2, provide high-frequency pose updates, relying solely on proprioceptive dead reckoning inevitably leads to unbounded, compounding drift due to wheel slip, mechanical tolerances, and discretization errors [1]. To address this limitation, this report presents the design and implementation of an EKF [2] to fuse additional external sensor readings. Fusing high-frequency kinematic predictions with absolute AprilTag measurements [3] relative to the world frame yields an optimal, bounded estimate of the planar state (x, y, θ).

This work details the complete localization pipeline, from monocular camera calibration, measurement model noise covariance modeling to offline and online pose estimation evaluations. Offline experiments also specifically validate the filter's rapid convergence from degraded initial conditions and its resilience to intermittent visual occlusions. Finally, the estimator's real-time capabilities are demonstrated through a custom `NiceGUI` web dashboard. Within this framework, the EKF serves as the backbone for a closed-loop Proportional-Derivative (PD) controller [4], enabling the Ackermann robot to autonomously navigate to target waypoints while actively rejecting odometry drift.

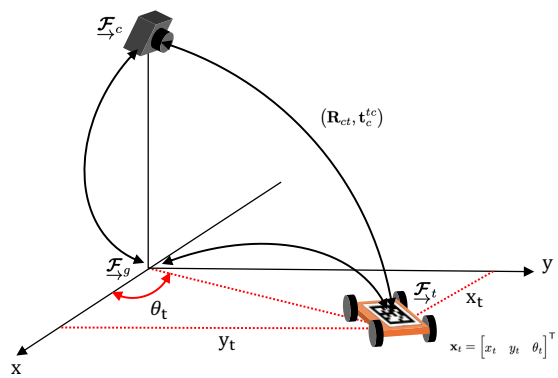

Figure 1: Coordinate frames and measurement model. The global world frame $\underrightarrow{\mathcal{F}}_g$ is fixed to the environment, and the camera frame $\underrightarrow{\mathcal{F}}_c$ is stationary. The AprilTag attached to the robot defines the tag frame $\underrightarrow{\mathcal{F}}_t$. The relative pose $(\mathbf{R}_{ct}, \mathbf{t}_c^{tc})$ is estimated in the camera frame and transformed into the world frame to obtain the planar state $\mathbf{x}_t = [x_t \; y_t \; \theta_t]^\mathsf{T}$.

II. Method

A. Notation

In this report, Latin and Greek letters, such as a and α, represent scalar variables. Lowercase (e.g., $\mathbf{h}$ and $\boldsymbol{\sigma}$) and uppercase (e.g., $\boldsymbol{\Theta}$ and $\mathbf{C}$) boldface characters are reserved for vectors and matrices, respectively. A Cartesian reference frame is identified by $\underrightarrow{\mathcal{F}}_{\cdot}$. The translation vector from $\underrightarrow{\mathcal{F}}_a$ to $\underrightarrow{\mathcal{F}}_b$, expressed in $\underrightarrow{\mathcal{F}}_a$, is denoted by $\mathbf{t}_a^{ba}$. The function $\mathbf{R}(\theta)$ maps $\theta \in \mathbb{R}$ to an element of $\mathrm{SO}(2)$; for example, $\mathbf{R}(\theta_{ab})$ defines the rotation from $\underrightarrow{\mathcal{F}}_b$ to $\underrightarrow{\mathcal{F}}_a$.

For the EKF formulation [2], $\check{(\cdot)}$ denotes a predicted (a priori) quantity, while $\hat{(\cdot)}$ denotes an updated (a posteriori) estimate. For example, $\check{\mathbf{x}}_t$ and $\check{\mathbf{P}}_t$ represent the predicted state mean and covariance, and $\hat{\mathbf{x}}_t$ and $\hat{\mathbf{P}}_t$ represent the corrected estimates after incorporating measurements.

B. System Model

The robot pose at time t, expressed in the global frame $\underrightarrow{\mathcal{F}}_g$, is

$$\mathbf{x}_t = \begin{bmatrix} x_t & y_t & \theta_t \end{bmatrix}^\mathsf{T} \in \mathbb{R}^3, \tag{1}$$

where $x_t, y_t \in \mathbb{R}$ denote planar position and $\theta_t \in \mathbb{R}$ denotes heading.

The control input is defined as the commanded translational speed and steering angle,

$$\mathbf{u}_t = \begin{bmatrix} v_t & \alpha_t \end{bmatrix}^\mathsf{T}, \tag{2}$$

where $v_t \in \mathbb{R}$ is the forward velocity command and $\alpha_t \in \mathbb{R}$ is the steering command.

From lab 2, on the hardware level, the commanded velocity is realized through wheel encoder measurements. The traveled distance over one sampling interval is computed as

$$s_t = v_t \Delta t, \tag{3}$$

or equivalently from encoder counts using a calibrated encoder constant.

Due to steering nonlinearities, the effective steering angle is modeled as

$$\delta_t = c_0 \alpha_t^2 + c_1 \alpha_t + c_2, \tag{4}$$

where $c_i \in \mathbb{R}$ are experimentally calibrated coefficients.

*a) **Motion Model**:* The discrete-time Ackermann model is

$$\mathbf{x}_t = \begin{bmatrix} x_{t-1} + s_t \cos\theta_{t-1} \\ y_{t-1} + s_t \sin\theta_{t-1} \\ \theta_{t-1} - \dfrac{s_t}{L} \tan\delta_t \end{bmatrix} + \mathbf{w}_t, \tag{5}$$

where $L \in \mathbb{R}$ is the wheelbase and

$$\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t). \tag{6}$$

The process covariance is defined as

$$\mathbf{R}_t = \begin{bmatrix} \sigma_v^2 & 0 & 0 \\ 0 & \sigma_v^2 & 0 \\ 0 & 0 & \sigma_\delta^2 \end{bmatrix}. \tag{7}$$

*b) **Measurement Model**:* The camera directly observes the global pose:

$$\mathbf{z}_t = \mathbf{x}_t + \mathbf{n}_t, \tag{8}$$

where

$$\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{9}$$

and

$$\mathbf{Q} = \begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_\theta^2 \end{bmatrix}. \tag{10}$$

C. Iterative EKF

The nonlinear system is modeled as

$$\mathbf{x}_t = g(\mathbf{x}_{t-1}, \mathbf{u}_t) + \mathbf{w}_t, \tag{11}$$

$$\mathbf{z}_t = h(\mathbf{x}_t) + \mathbf{n}_t, \tag{12}$$

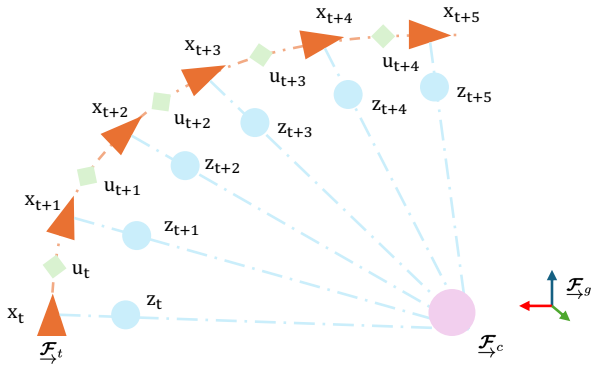

Figure 2: Factor graph representation of the robot state estimation and sensor fusion process. The robot's temporal poses, denoted by states $\mathbf{x_t}$ (orange triangles) in the local frame $\mathcal{F}_t$, evolve continuously according to control inputs $\mathbf{u_t}$ (green squares) driven by the dead reckoning kinematic model. A stationary observer, representing the camera frame $\mathcal{F}_c$, captures discrete visual measurements $\mathbf{z_t}$ (blue circles) of the robot to correct odometry drift. The system maps all poses and sensor data relative to the global coordinate frame $\mathcal{F}_g$.

where

$$\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \tag{13}$$

$$\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}). \tag{14}$$

*a) **Prediction Step**:* The motion model used in the implementation is

$$g(\mathbf{x}_{t-1}, \mathbf{u}_t) = \begin{bmatrix} x_{t-1} + s_t \cos\theta_{t-1} \\ y_{t-1} + s_t \sin\theta_{t-1} \\ \theta_{t-1} - \dfrac{s_t}{L} \tan\delta_t \end{bmatrix}, \tag{15}$$

where $s_t = K_e \Delta e_t$ and δ_t is the calibrated steering angle.

The predicted state and covariance are

$$\check{\mathbf{x}}_t = g(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t), \tag{16}$$

$$\check{\mathbf{P}}_t = \mathbf{F}_{t-1} \hat{\mathbf{P}}_{t-1} \mathbf{F}_{t-1}^\mathsf{T} + \mathbf{R}_t. \tag{17}$$

The Jacobian is defined as

$$\mathbf{F}_{t-1} = \frac{\partial g}{\partial \mathbf{x}_{t-1}} = \begin{bmatrix} 1 & 0 & -s_t \sin\theta_{t-1} \\ 0 & 1 & s_t \cos\theta_{t-1} \\ 0 & 0 & 1 \end{bmatrix}. \tag{18}$$

*b) **Correction Step**:* The measurement model directly observes the pose:

$$h(\mathbf{x}_t) = \mathbf{x}_t. \tag{19}$$

The innovation is

$$\tilde{\mathbf{y}}_t = \mathbf{z}_t - h(\check{\mathbf{x}}_t). \tag{20}$$

Since the measurement is linear, the measurement Jacobian is

$$\mathbf{G}_t = \frac{\partial h}{\partial \mathbf{x}_t} = \mathbf{I}. \tag{21}$$

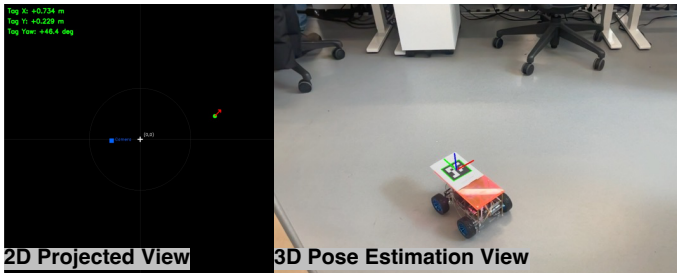

Figure 3: Real-time AprilTag 6-DoF pose estimation with world-frame projection. Left: 2D top-down map where the blue square denotes the fixed camera position, the red dot the tag position in the world frame, the green arrow the tag heading (yaw, +Y axis), and the white cross the initialized world origin. Right: 3D camera-view pose estimation with coordinate axes overlay.

The Kalman gain is

$$\mathbf{K}_t = \check{\mathbf{P}}_t \mathbf{G}_t^\top \left(\mathbf{G}_t \check{\mathbf{P}}_t \mathbf{G}_t^\top + \mathbf{Q} \right)^{-1}. \tag{22}$$

The posterior update is

$$\hat{\mathbf{x}}_t = \check{\mathbf{x}}_t + \mathbf{K}_t \tilde{\mathbf{y}}_t, \tag{23}$$

$$\hat{\mathbf{P}}_t = (\mathbf{I} - \mathbf{K}_t \mathbf{G}_t) \check{\mathbf{P}}_t. \tag{24}$$

Angular innovations are wrapped to $(-\pi, \pi]$ to ensure consistency.

III. Camera-Based Pose Measurement Model

A video camera is mounted on the side of the experimental scene, as shown in Figure 4. The camera is positioned approximately $45°$ downward to maximize coverage of the workspace while maintaining sufficient resolution. An AprilTag is rigidly attached to the top surface of the mobile robot to provide global pose measurements for the EKF measurement model.

A. Camera Intrinsics and Distortion Correction

The camera is modeled using the standard pinhole projection model [5], which maps a 3D point in the camera frame onto the image plane. The intrinsic calibration matrix is defined as

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{25}$$

where (f_x, f_y) denote the focal lengths in pixel units and (c_x, c_y) represent the principal point.

Lens distortion is modeled using a distortion parameter vector

$$\boldsymbol{d} = \begin{bmatrix} k_1 & k_2 & p_1 & p_2 & k_3 \end{bmatrix}^\top, \tag{26}$$

where (k_1, k_2, k_3) are radial distortion coefficients and (p_1, p_2) are tangential distortion coefficients. Calibration is accomplished utilizing the OpenCV library [6].

B. AprilTag Detection and 3D Pose Recovery

An AprilTag from the 25h9 family (ID 5) [3] with side length $s_{\text{tag}} = 3$ inches is rigidly attached to the top of the mobile robot and used as the exteroceptive measurement source. The tag is printed on A4 paper and detected in each incoming frame.

Given calibrated camera parameters $(\mathbf{K}, \boldsymbol{d})$, images are first undistorted. The four detected pixel corners $\{\mathbf{p}_i\}_{i=1}^4$ are associated with the known 3D corner coordinates in the tag frame $\underrightarrow{\mathcal{F}}_t$:

$$\mathbf{P}_i = \begin{bmatrix} -\frac{s_{\text{tag}}}{2} & \frac{s_{\text{tag}}}{2} & 0 \\ \frac{s_{\text{tag}}}{2} & \frac{s_{\text{tag}}}{2} & 0 \\ \frac{s_{\text{tag}}}{2} & -\frac{s_{\text{tag}}}{2} & 0 \\ -\frac{s_{\text{tag}}}{2} & -\frac{s_{\text{tag}}}{2} & 0 \end{bmatrix}. \tag{27}$$

The tag pose relative to the camera frame $\underrightarrow{\mathcal{F}}_c$ is obtained by solving a Perspective-n-Point (PnP) problem [6] that minimizes reprojection error:

$$\min_{\mathbf{R}_{ct}, \mathbf{t}_c^{tc}} \sum_{i=1}^{4} \left\| \mathbf{p}_i - \pi(\mathbf{K}, \mathbf{R}_{ct}, \mathbf{t}_c^{tc}, \mathbf{P}_i) \right\|^2. \tag{28}$$

The solution yields the rigid transformation $(\mathbf{R}_{ct}, \mathbf{t}_c^{tc})$, where $\mathbf{R}_{ct} \in \mathrm{SO}(3)$ and $\mathbf{t}_c^{tc} \in \mathbb{R}^3$.

C. Extrinsic Calibration

A global world frame $\underrightarrow{\mathcal{F}}_g$ is defined as a fixed reference frame attached to the environment. The camera is assumed stationary during the experiment; therefore, the camera frame $\underrightarrow{\mathcal{F}}_c$ is time-invariant.

At initialization, the robot is placed near the designated origin on the floor and aligned with the world axes. The observed AprilTag pose at this instant, $(\mathbf{R}_{ct,0}, \mathbf{t}_{c,0}^{tc})$, is defined as the world-frame origin. This initialization establishes the rigid transformation between the world and camera frames:

$$\mathbf{R}_{gc} = \mathbf{R}_{ct,0}^\top, \qquad \mathbf{t}_g^{cg} = -\mathbf{R}_{ct,0}^\top \mathbf{t}_{c,0}^{tc}. \tag{29}$$

Thus, $\underrightarrow{\mathcal{F}}_g$ is defined relative to the fixed camera frame using the initial tag measurement.

For subsequent frames, the AprilTag pose $(\mathbf{R}_{ct}, \mathbf{t}_c^{tc})$ is estimated in the camera frame. The tag position expressed in the world frame is computed as

$$\mathbf{p}_g^t = \mathbf{R}_{ct,0}^\top \left(\mathbf{t}_c^{tc} - \mathbf{t}_{c,0}^{tc} \right). \tag{30}$$

This transformation removes the initial offset and rotates the measurement into the world frame, ensuring a consistent global coordinate representation for the mobile robot.

Finally, since the robot motion is constrained to the ground plane, the 3D world-frame pose is reduced to its planar 2D components. Let $\mathbf{p}_g^t = \begin{bmatrix} x_g^t & y_g^t & z_g^t \end{bmatrix}^\top$. The planar position is obtained by projection onto the x–y plane,

$$\begin{bmatrix} x_g^t \\ y_g^t \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{p}_g^t, \tag{31}$$

and the heading is extracted from the world-frame rotation as $\theta_g^t = \operatorname{atan2}\left((\mathbf{R}_{gt})_{21}, (\mathbf{R}_{gt})_{11}\right)$.

The EKF correction measurement is therefore

$$\mathbf{z}_t = \begin{bmatrix} x_g^t \\ y_g^t \\ \theta_g^t \end{bmatrix}. \tag{32}$$

D. Measurement Model Covariance Estimation

To determine a realistic measurement covariance $\mathbf{Q}$, an empirical calibration procedure is performed. The robot is placed at N known ground-truth poses $\{(x^{(i)}, y^{(i)}, \theta^{(i)})\}_{i=1}^N$ in the world frame $\underrightarrow{\mathcal{F}}_g$, measured using tape markers (see Figure 4). For each placement, the corresponding camera-based pose estimate $\mathbf{z}^{(i)} = \begin{bmatrix} z_x^{(i)} & z_y^{(i)} & z_\theta^{(i)} \end{bmatrix}^\mathsf{T}$ is recorded.

The measurement residual for trial i is computed as

$$\mathbf{e}^{(i)} = \begin{bmatrix} z_x^{(i)} - x^{(i)} \\ z_y^{(i)} - y^{(i)} \\ \operatorname{wrap}\left(z_\theta^{(i)} - \theta^{(i)}\right) \end{bmatrix}, \tag{33}$$

where the angular component is wrapped to $(-\pi, \pi]$ to respect periodicity.

Component-wise sample variances are then computed using an unbiased estimator:

$$\sigma_i^2 = \frac{1}{N-1} \sum_{k=1}^{N} \left(e_i^{(k)} - \bar{e}_i\right)^2, \quad i \in \{x, y, \theta\}, \tag{34}$$

where $\bar{e}_i$ denotes the empirical mean of the corresponding residual component.

The EKF measurement covariance is constructed as

$$\mathbf{Q} = \operatorname{diag}\left\{\sigma_x^2, \sigma_y^2, \sigma_\theta^2\right\}, \tag{35}$$

which reflects independent noise assumptions in position and heading. Table I summarizes the ground-truth poses and corresponding camera measurements used for identification.

Table I: System Identification Poses and Camera Estimates

Trial	Ground Truth (x, y, θ)	Measurement (z_x, z_y, z_θ)
1	$(0.00, 0.00, 1.57)$	$(0.000, 0.000, 1.571)$
2	$(0.00, 0.50, 1.57)$	$(0.001, 0.471, 1.539)$
3	$(0.00, 1.00, 1.57)$	$(0.018, 0.964, 1.549)$
4	$(0.00, 0.50, 3.14)$	$(-0.009, 0.470, 3.108)$
5	$(0.00, 0.50, 4.71)$	$(0.008, 0.462, 4.702)$
6	$(0.00, 0.50, 1.57)$	$(0.007, 0.469, 1.547)$
7	$(0.50, 0.50, 1.57)$	$(0.639, 0.702, 1.533)$
8	$(-0.50, 0.50, 1.57)$	$(-0.581, 0.793, 1.527)$

IV. Offline Experiments

A. Data Collection

All experiments were conducted in the same indoor planar environment used during the measurement model calibration. The complete setup is illustrated in Figure 4.

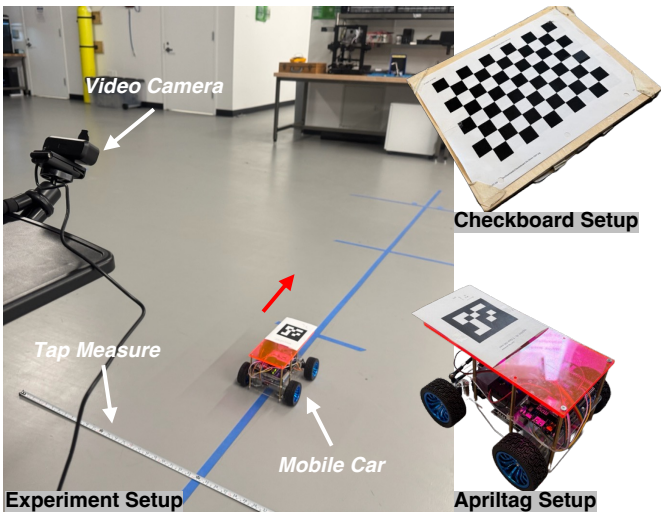

Figure 4: Experimental setup for camera-based localization. Left: Video camera mounted at approximately 45° provides global AprilTag-based pose measurements of the mobile robot; tape markers define ground-truth positions in the world frame. Top-right: Checkerboard used for intrinsic camera calibration. Bottom-right: AprilTag (ID 5, 3-inch) rigidly mounted on the robot for 6-DoF pose estimation.

The offline datasets were generated by executing controlled trials through the custom GUI, which simultaneously logged actuator commands and sensor outputs. For each timestep, the system recorded wheel encoder–derived velocity, steering commands, and the corresponding AprilTag-based pose estimate. All signals were stored with synchronized timestamps to preserve temporal consistency between motion inputs and measurements. In parallel, the raw RGB video stream from the webcam was archived to enable verification of tag detection quality and offline debugging.

To validate the EKF performance and the dataset collected, three experiments are conducted: 1) EKF estimation test, 2) noisy initial guess, and 3) occlusion tests.

B. Offline Iterative EKF

The iterative EKF was evaluated across all offline-collected datasets. During physical experiments, factors such as robot speed, aggressive steering maneuvers, and workspace boundaries occasionally caused the AprilTag to become undetectable by the overhead camera. During these occlusion events, the EKF correction step is bypassed, and the state covariance is propagated purely via the dead-reckoning motion model.

Figure 5 illustrates a nominal experimental run where the camera maintained continuous line-of-sight with the robot. In this scenario, the iterative EKF consistently fuses global measurements, maintaining near-zero state estimation errors and tightly bounded 3σ confidence intervals throughout the trajectory.

Conversely, Figure 6 depicts an experiment where the camera completely lost track of the AprilTag just seconds into the run. Without exteroceptive corrections, the filter defaults to open-loop dead reckoning. As expected, the state estimate

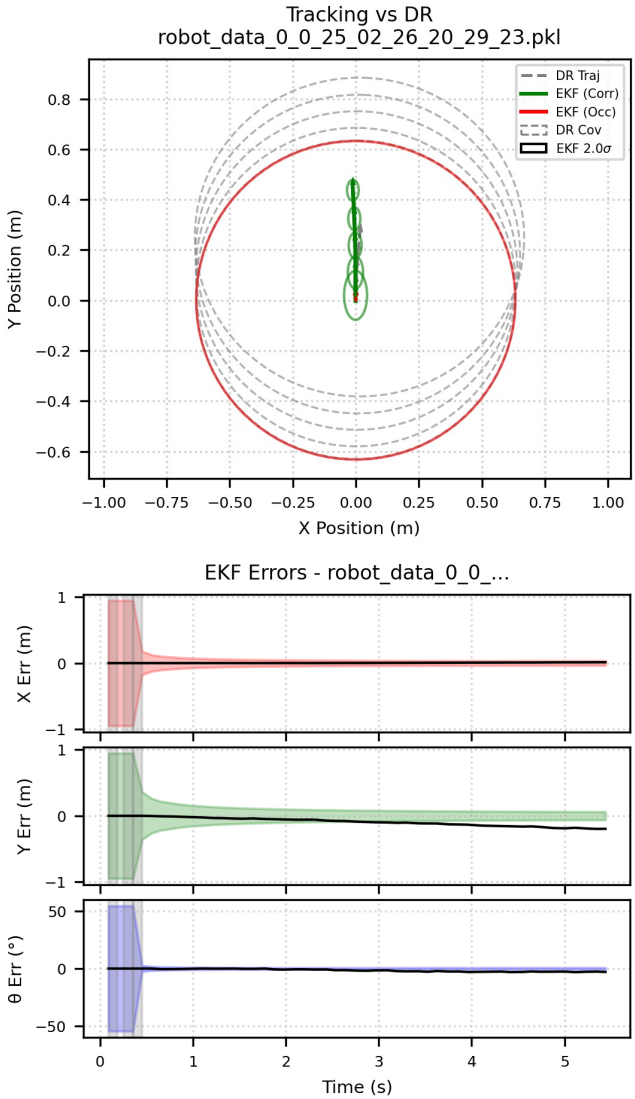

Figure 5: EKF tracking performance under ideal conditions with continuous camera visibility. The EKF trajectory (green) tightly tracks the ground truth, keeping positional and heading errors bounded near zero with narrow $\pm 3\sigma$ confidence envelopes.

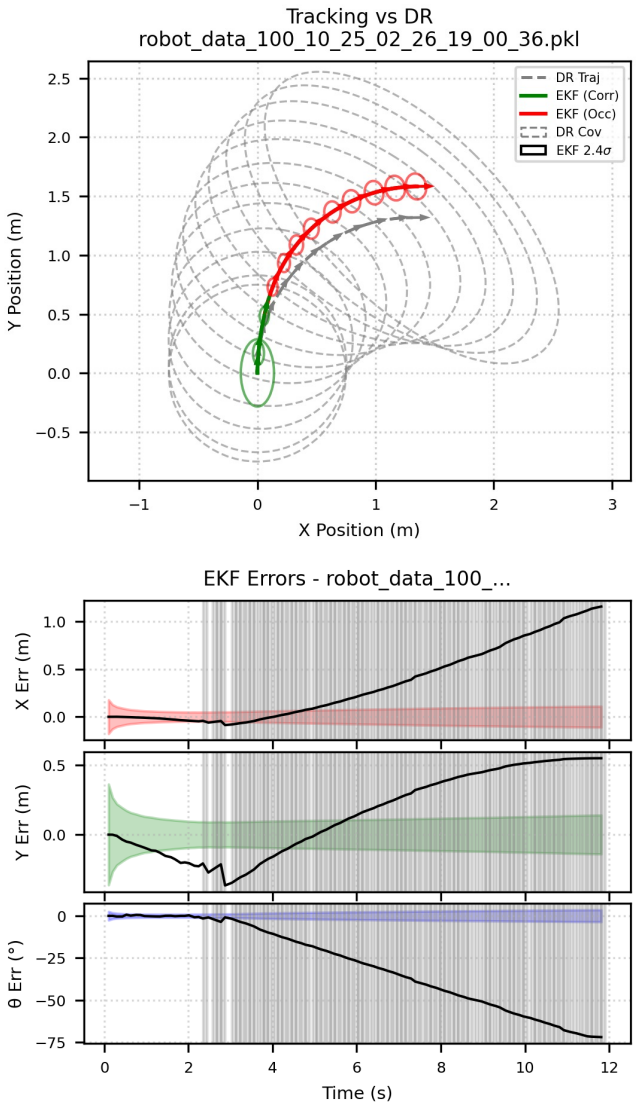

Figure 6: EKF tracking performance during severe, sustained camera occlusion. After a brief period of successful tracking (green), the camera loses the AprilTag (red trajectory, gray shaded error regions). The system is forced to rely entirely on dead reckoning, resulting in unbounded error accumulation and massive covariance inflation.

diverges rapidly from the true trajectory, and the uncorrected covariance bounds expand massively. These comparative results clearly demonstrate that while the kinematic model provides smooth high-frequency predictions [7], continuous visual feedback is strictly required to bound the cumulative drift inherent to the mobile robot.

C. Occlusion Test

To evaluate the filter's robustness to measurement loss, artificial periodic occlusions were introduced during post-processing. A periodic occlusion mask $\mathcal{O}(t)$ is defined as:

$$\mathcal{O}(t) = \begin{cases} 1, & \text{if } (t \bmod 1.5) > 1.0, \\ 0, & \text{otherwise.} \end{cases} \tag{36}$$

This creates a $1.5\,\mathrm{s}$ cycle comprising $1.0\,\mathrm{s}$ of visibility and $0.5\,\mathrm{s}$ of total occlusion. When $\mathcal{O}(t) = 1$, the camera measurement is forcibly removed ($\mathbf{z}_t = \varnothing$), forcing the filter to rely entirely on the kinematic prediction step.

The results are presented in Figure 7. During visible phases (green trajectory), the EKF actively fuses measurements to keep errors near zero. During occlusions (red trajectory, gray shaded regions), the filter propagates solely via the motion model, and the $\pm 3\sigma$ confidence envelopes appropriately expand to reflect the accumulating uncertainty. Ultimately, the EKF successfully restricts the severe, unbounded drift inherent to the open-loop dead reckoning (dashed gray), confirming its reliability in navigating through intermittent sensor dropouts.

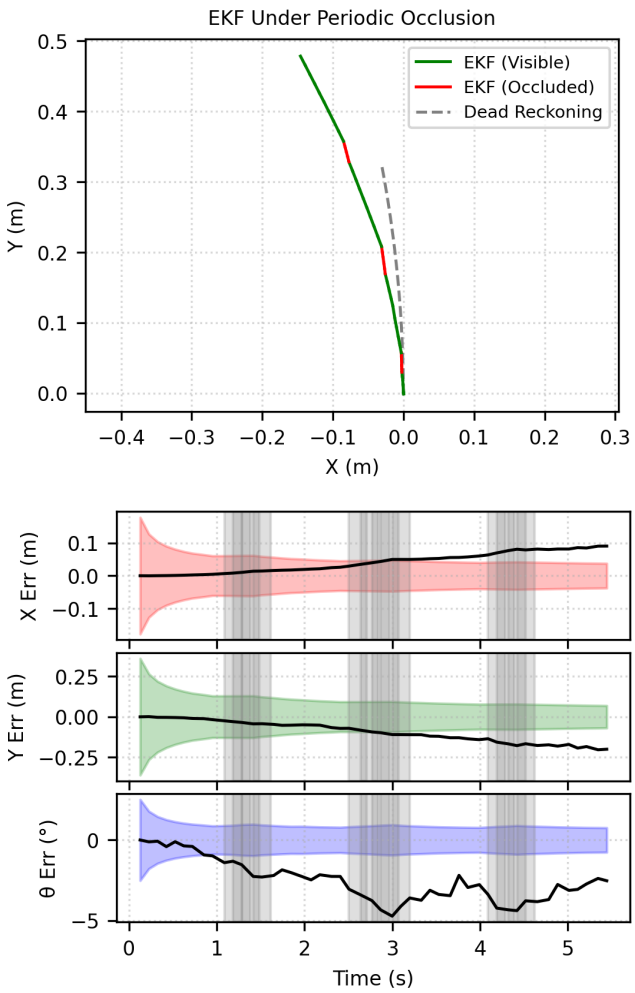

Figure 7: EKF performance under periodic occlusions. Top: Trajectory comparison showing the EKF (green/red) successfully mitigating the severe drift of pure dead reckoning (dashed gray). Bottom: State estimation errors with $\pm 3\sigma$ bounds. Gray shaded regions indicate $0.5\,\mathrm{s}$ intervals of simulated measurement loss.

D. Initial Condition Robustness

To evaluate EKF's robustness to bad initialization, artificial position offsets ($0.25\,\mathrm{m}$ to $2.0\,\mathrm{m}$) are injected into the initial state $\mathbf{x}_0$. The initial covariance is correspondingly inflated to prioritize incoming camera corrections during the transient phase.

The convergence behavior over the first $1.0\,\mathrm{s}$ is shown in Figure 8. The EKF rapidly eliminates the injected offsets, with all error norms converging to a shared baseline within $0.5\,\mathrm{s}$. While a nominal tracking error emerges subsequently, it affects all trajectories equally. This demonstrates the filter's robust convergence capabilities, successfully recovering the true robot pose independent of the initial perturbation magnitude.

V. Online Experiments

A. Online EKF

To evaluate the practicality of the EKF in real-time, the filter was integrated into the robot's control dashboard via

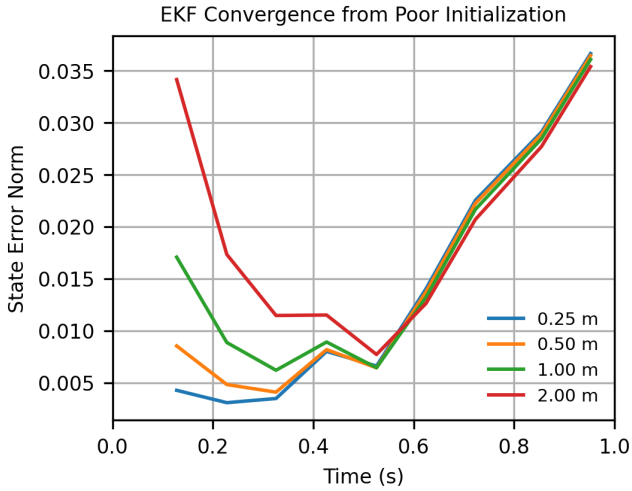

Figure 8: EKF state error norm convergence. The filter rapidly corrects initial state errors of varying magnitudes, converging to a shared baseline tracking error within $0.5\,\mathrm{s}$.

a `NiceGUI` web interface used in lab2. This GUI simultaneously manages the asynchronous video stream from the overhead camera, the UDP telemetry link with the Arduino hardware, and the recursive state estimation loop.

During online validation, the system executes the EKF at approximately $10\,\mathrm{Hz}$. In each cycle, the robot polls the latest wheel encoder counts and steering commands ($\mathbf{u}_t$) to perform the prediction step. And the latest camera frame is processed to extract the AprilTag pose ($\mathbf{z}_t$). The control loop actively checks for measurement occlusions; if the AprilTag is undetected or the camera signal drops, the filter gracefully falls back to dead reckoning without interrupting the robot's operation.

The resulting estimated state ($\mathbf{x}_t$), along with its dynamically scaled 3σ covariance ellipse, is rendered in real-time on the dashboard alongside the uncorrected dead reckoning trajectory. This online deployment confirms that the EKF implementation is computationally efficient enough to provide stable, real-time sensor fusion and drift correction for closed-loop navigation.

B. Online Closed-Loop PD Control

To further validate the performance of the EKF state estimator, a Proportional-Derivative (PD) controller was designed to operate in tandem with the localization system [4]. This integration enables the mobile robot to autonomously navigate from an initial position to a user-defined target waypoint.

The controller maps the error between the current EKF-estimated state $\hat{\mathbf{x}}_t = [\hat{x}_t, \hat{y}_t, \hat{\theta}_t]^\mathsf{T}$ and the desired goal pose $\mathbf{x}_g = [x_g, y_g, \theta_g]^\mathsf{T}$ into actionable actuator commands $\mathbf{u}_t = [v_t, \alpha_t]^\mathsf{T}$. The control objective is decomposed into translational and rotational components. The Euclidean distance error $e_{d,t}$ is defined as:

$$e_{d,t} = \sqrt{(x_g - \hat{x}_t)^2 + (y_g - \hat{y}_t)^2}. \tag{37}$$

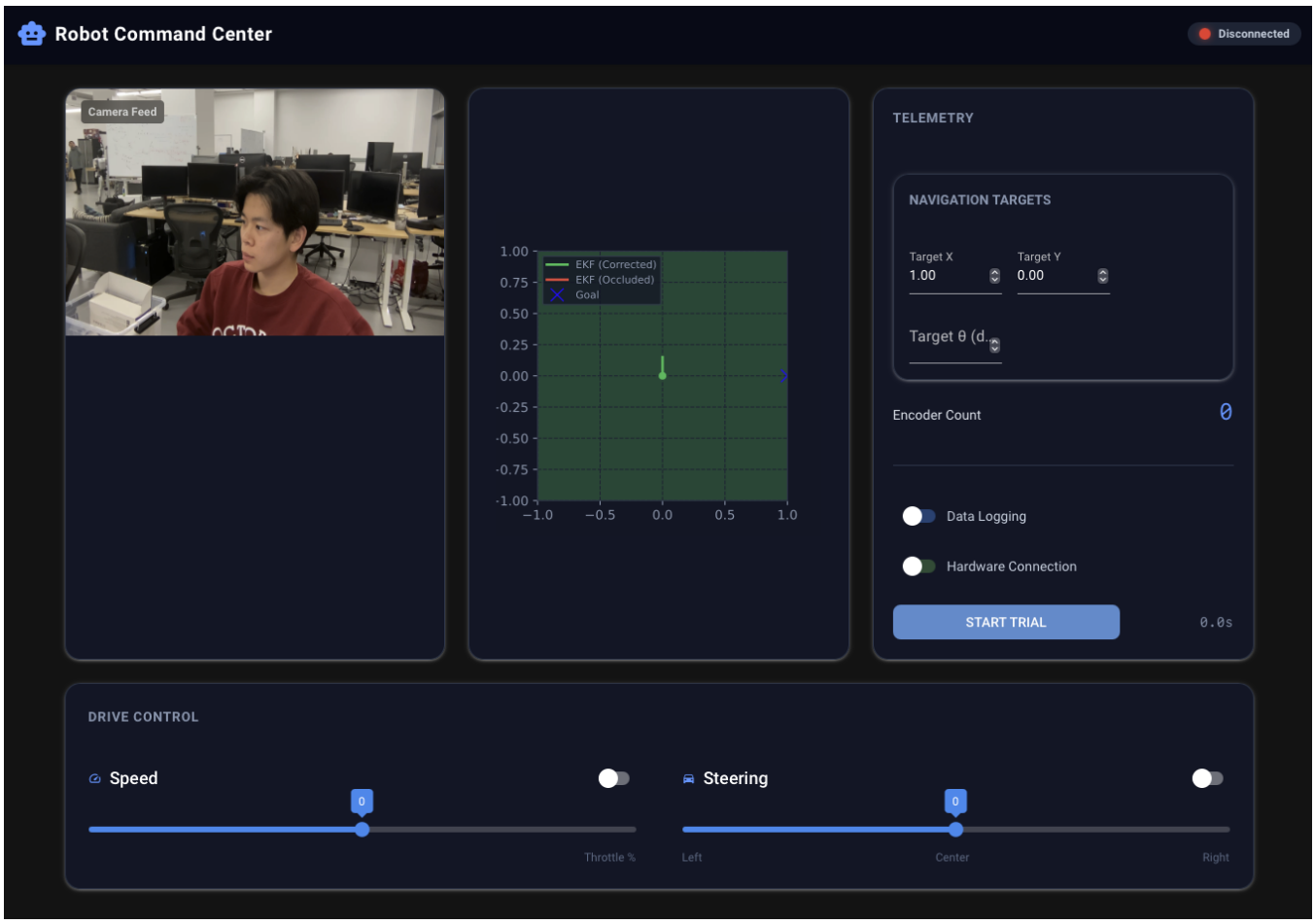

Figure 9: The updated NiceGUI interface displaying live video feed, real-time EKF trajectory map, hardware telemetry, and the user input module for desired global targets.

For steering control, the heading error $e_{\theta,t}$ toward the target is calculated as:

$$e_{\theta,t} = \operatorname{atan2}(y_g - \hat{y}_t, x_g - \hat{x}_t) - \hat{\theta}_t. \tag{38}$$

To ensure the controller selects the shortest angular path, the error is normalized to the range $(-\pi, \pi]$ using a standard wrapping function:

$$e_{\theta,t} \leftarrow \operatorname{atan2}(\sin(e_{\theta,t}), \cos(e_{\theta,t})). \tag{39}$$

The actuator commands for linear velocity v_t and steering angle α_t are derived using the following PD control laws:

$$v_t = K_{p,v} e_{d,t} + K_{d,v} \frac{e_{d,t} - e_{d,t-1}}{\Delta t}, \tag{40}$$

$$\alpha_t = K_{p,\alpha} e_{\theta,t} + K_{d,\alpha} \frac{e_{\theta,t} - e_{\theta,t-1}}{\Delta t}, \tag{41}$$

where K_p represents the proportional gain governing the response speed, and K_d represents the derivative gain used to dampen oscillations and mitigate overshoot as the robot approaches the target pose [4].

To facilitate user interaction, the web-based GUI (Figure 9) was updated to include a dedicated navigation module. Through this interface, the user can directly input the desired target pose (x_g, y_g, θ_g). Upon initiating a trial, the system engages the closed-loop PD controller, utilizing the real-time EKF state estimates to autonomously navigate the robot to the specified destination.

Despite the successful integration of the EKF and PD controller within the web-based dashboard, physical real-world validation could not be completed prior to submission due to an unforeseen hardware failure related to the robot's power supply. The physical deployment and experimental tuning of the autonomous navigation framework are deferred to future work once the hardware issues are resolved.

VI. Conclusion

The implemented EKF provides robust planar localization by successfully bounding the Ackermann kinematic model's inherent drift. Offline evaluations demonstrated the filter's rapid convergence—recovering from initial state errors of up to $2.0\,\mathrm{m}$ within $0.5\,\mathrm{s}$—and its resilience to visual occlusions by accurately propagating motion uncertainty until measurements resumed. While the EKF was successfully integrated into an asynchronous, real-time framework for closed-loop PD navigation, a hardware failure prevented physical validation. Future work will focus on resolving the power supply issues and validate the autonomous navigation pipeline.

REFERENCES

[1] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005.

[2] T. D. Barfoot, *State Estimation for Robotics*. Cambridge University Press, 2017.

[3] J. Wang and E. Olson, "AprilTag 2: Efficient and robust fiducial detection," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4193–4198, IEEE, 2016.

[4] K. Ogata, *Modern Control Engineering*. Prentice Hall, 5th ed., 2010.

[5] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.

[6] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, vol. 25, no. 11, pp. 120–123, 2000.

[7] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*. MIT Press, 2011.